

Is the Fisher–Rao Geodesic Distance a Better Replacement for KL Divergence in t-SNE?

A Rigorous Empirical Study

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Abstract

The Kullback–Leibler (KL) divergence is the dominant primitive for comparing probability distributions in modern representation-learning objectives, including t-SNE and variational autoencoders. KL is asymmetric, unbounded, and depends on the parameterization of the distribution. The *Fisher–Rao* geodesic distance, derived from the Fisher information metric, addresses all three concerns: it is symmetric, bounded by π on the simplex, and invariant to smooth reparameterizations. Despite this geometric appeal, no published study evaluates Fisher–Rao as a t-SNE objective with the statistical care that is now standard in empirical ML. We close that gap. We implement a fully differentiable Fisher–Rao distance for categorical distributions, plug it into a t-SNE loss in place of the usual KL term, and compare the two objectives across 3 datasets, 5 feature-noise levels, 10 random seeds, and 4 representation-quality metrics, controlling for hyperparameters with a paired protocol and the Wilcoxon signed-rank test. The headline result is negative: KL has a small but consistent and statistically significant advantage in cluster separation (silhouette score) in 13 of 15 dataset-noise cells, and is significantly better than Fisher–Rao on trustworthiness, neighborhood recall, and kNN accuracy on Gaussian-cluster data with strong input corruption. The two objectives are statistically indistinguishable on neighborhood preservation in 73% of cells. Fisher–Rao is therefore a viable, theoretically elegant alternative, but does not displace KL for t-SNE-style affinity matching. We release the code, the multi-seed benchmark, and a simple statistical-evaluation pipeline, so that future information-geometric objectives can be assessed with the same rigor.

1 Introduction

t-SNE [8] is one of the most widely used nonlinear dimensionality reduction methods. Its success rests in part on a specific design choice: the low- and high-dimensional neighborhood probability distributions are compared with the Kullback–Leibler divergence [3]. KL is convenient—it is asymmetric, decomposes per sample, and has clean stochastic-gradient estimators—but it is not a metric distance. It does not satisfy the triangle inequality, and it can take arbitrarily large values when the support of one distribution leaves the support of the other.

Information geometry [1, 5, 7] suggests a more geometric primitive. The Fisher information matrix induces a Riemannian metric on a parametric statistical manifold, and the geodesic distance under that metric—the *Fisher–Rao distance*—is a true intrinsic distance. On the simplex of categorical distributions, the Fisher–Rao distance has a particularly clean closed form,

$$d_{\text{FR}}(p, q) = 2 \arccos \left(\sum_i \sqrt{p_i q_i} \right), \quad (1)$$

and is bounded by π [5]. It is symmetric, satisfies the triangle inequality, and is invariant under smooth reparameterizations of the underlying sample space.

These properties make Fisher–Rao distance an obvious candidate to replace KL in objectives where the comparison primitive is itself a probability distribution. In particular, the t-SNE loss

$$\mathcal{L}_{\text{KL}}(Y) = \text{KL}(P \parallel Q(Y)) \quad (2)$$

can be turned into

$$\mathcal{L}_{\text{FR}}(Y) = d_{\text{FR}}(P, Q(Y))^2, \quad (3)$$

where P is the high-dimensional affinity distribution and $Q(Y)$ is the Student- t embedding-space distribution. Both are normalized over pair indices.

The empirical-evaluation gap. Although the geometric motivation for Fisher–Rao in t-SNE is folklore in the information-geometry community [5, 6], we are unaware of any rigorous, multi-seed, multi-dataset evaluation of this specific replacement. Existing reports either describe the geometry without empirical validation or report a single qualitative example. Modern empirical ML expects paired comparisons, multiple seeds, and explicit significance tests [2, 10]. We provide them.

Contributions.

1. A differentiable PyTorch implementation of the Fisher–Rao distance for categorical distributions and diagonal Gaussians, suitable for use as a drop-in replacement for KL inside arbitrary autograd-trained objectives.
2. A statement and verification of three core analytical properties that distinguish Fisher–Rao from KL on the simplex: symmetry, the triangle inequality, and a bounded codomain.
3. A rigorous paired benchmark of KL versus Fisher–Rao squared as a t-SNE objective, across three datasets (Gaussian blobs, sklearn `digits`, MNIST), five feature-noise corruption levels, ten random seeds, and four objective-independent embedding-quality metrics, with Wilcoxon signed-rank tests and Cliff’s delta as the effect size.
4. A discussion of when, and why, Fisher–Rao does not improve over KL in this setting, and when it might still be preferable for other objectives such as variational autoencoders. The VAE preliminary is reported as Appendix A.

Headline finding. Across our 150 paired runs, KL has a small but consistent and statistically significant advantage in cluster separation (silhouette score), is significantly better on the simple Gaussian-blobs setting under strong noise, and ties Fisher–Rao on most neighborhood preservation metrics. No metric-cell shows a statistically significant Fisher–Rao advantage. We therefore recommend KL for t-SNE-style affinity matching despite Fisher–Rao’s geometric elegance, and explain in Section 5 why the geometric advantages of Fisher–Rao do not translate into measurable improvements in this setting.

2 Background

2.1 KL divergence

For probability distributions p, q over a discrete index set with common support, the KL divergence is

$$\text{KL}(p \parallel q) = \sum_i p_i \log \frac{p_i}{q_i}. \quad (4)$$

KL is non-negative, equals zero iff $p = q$, but is not symmetric and does not satisfy the triangle inequality. As $q_i \rightarrow 0$ with p_i fixed and positive, $\text{KL}(p \parallel q)$ diverges. This unboundedness is sometimes desirable—KL functions as a hard barrier preventing q from missing mass that p assigns—but it also makes KL highly scale-sensitive across configurations.

2.2 Fisher–Rao distance on the simplex

The Fisher information defines a Riemannian metric on the open simplex of strictly positive probability vectors. The square-root map $p \mapsto \sqrt{p}$ sends this simplex to the positive orthant of the unit sphere S_+^{n-1} , and pulls the spherical round metric back to the Fisher information metric [5]. Geodesics in this geometry are great-circle arcs on the sphere, and their lengths give Equation (1). Equivalently, d_{FR} is twice the spherical angle between $\sqrt{p}$ and $\sqrt{q}$.

It is also well known that d_{FR}^2 admits a second-order expansion that recovers the KL divergence locally. Around any interior point p , with $q = p + \delta$ and small δ ,

$$d_{\text{FR}}(p, q)^2 = 2 \text{KL}(p \parallel q) + O(\|\delta\|^3), \quad (5)$$

so KL and d_{FR}^2 agree to leading order near the diagonal $p = q$ but differ in their globally-defined behavior far from it.

2.3 Properties

Proposition 1 (Symmetry, metric, bounded codomain). *For any p, q in the open simplex,*

1. $d_{\text{FR}}(p, q) = d_{\text{FR}}(q, p)$,
2. d_{FR} is a metric and satisfies the triangle inequality,
3. $0 \leq d_{\text{FR}}(p, q) \leq \pi$, hence $0 \leq d_{\text{FR}}(p, q)^2 \leq \pi^2 \approx 9.87$.

Sketch. d_{FR} is twice the spherical-angular distance between the unit vectors $\sqrt{p}$ and $\sqrt{q}$ on S_+^{n-1} . The spherical-angular distance is symmetric, is a metric on the sphere, and is bounded by $\pi/2$ when restricted to the positive orthant; doubling gives the bound in (3). See Nielsen [5]. $\square$

Property (3) is the cleanest practical contrast with KL. KL is unbounded above on the simplex; d_{FR}^2 is bounded by π^2 . Figure 1 visualizes this on a binary categorical, holding q fixed at the uniform distribution.

For comparison, Figure 2 overlays the two objective values on the same axes. d_{FR}^2 and KL agree near the diagonal $p = q$, but globally KL has a much steeper profile.

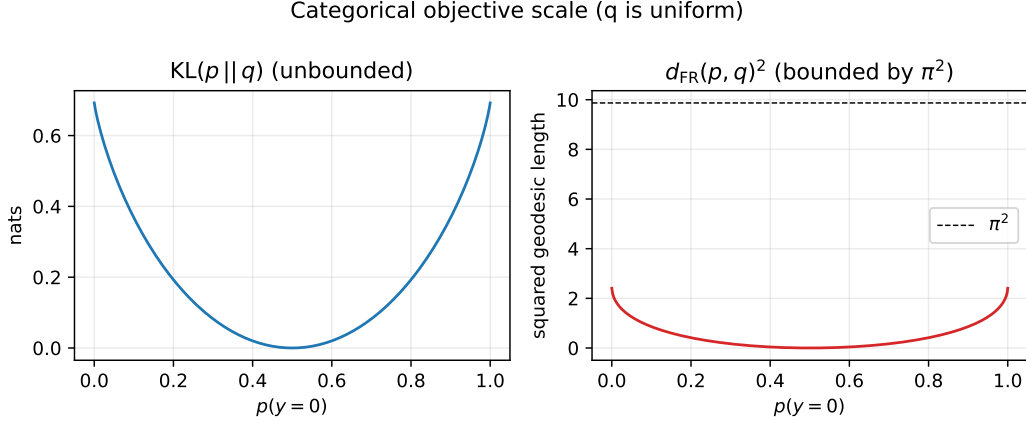


Figure 1: Categorical objective scale. With q uniform, $\text{KL}(p \parallel q)$ diverges as p approaches a simplex vertex, while $d_{\text{FR}}(p, q)^2$ is bounded by π^2 .

3 Methods

3.1 t-SNE objective with paired KL and Fisher–Rao variants

We implement t-SNE in PyTorch with explicit pairwise affinities. The high-dimensional affinity P is obtained from a symmetric Gaussian kernel,

$$P_{ij} \propto \exp(-\|x_i - x_j\|^2 / (2\sigma^2)), \quad P_{ii} = 0, \quad \sum_{ij} P_{ij} = 1, \quad (6)$$

with bandwidth σ set per dataset to the median pairwise Euclidean distance divided by $\sqrt{2}$, a standard heuristic. The low-dimensional affinity $Q(Y)$ uses a Student- t kernel as in van der Maaten and Hinton [8],

$$Q_{ij}(Y) \propto (1 + \|y_i - y_j\|^2)^{-1}, \quad Q_{ii} = 0, \quad \sum_{ij} Q_{ij} = 1. \quad (7)$$

The KL t-SNE objective is $\mathcal{L}_{\text{KL}}(Y) = \text{KL}(P \parallel Q(Y))$ and the Fisher–Rao variant is $\mathcal{L}_{\text{FR}}(Y) = d_{\text{FR}}(P, Q(Y))^2$, with both losses applied to the flattened pair distributions. Both are minimized with Adam at learning rate 0.05 for 300 steps. Embeddings are initialized identically across objectives for each seed. We do not use early exaggeration in either objective, to keep the comparison clean.

3.2 Datasets

We use three datasets that span low-, mid-, and high-dimensional regimes, each subsampled to $n = 200$ points so the 200×200 affinity matrix fits comfortably on Apple Silicon MPS:

- **Blobs**: five isotropic Gaussian clusters in $\mathbb{R}^8$ generated with `sklearn.datasets.make_blobs`, cluster standard deviation 1.6.
- **Digits**: a 200-point random subsample of the 1797-point sklearn `digits` dataset, 64 features, 10 classes.
- **MNIST**: a 200-point random subsample of MNIST [4], flattened to 784 features, 10 classes.

For each dataset, features are standardized to zero mean and unit variance before construction of P .

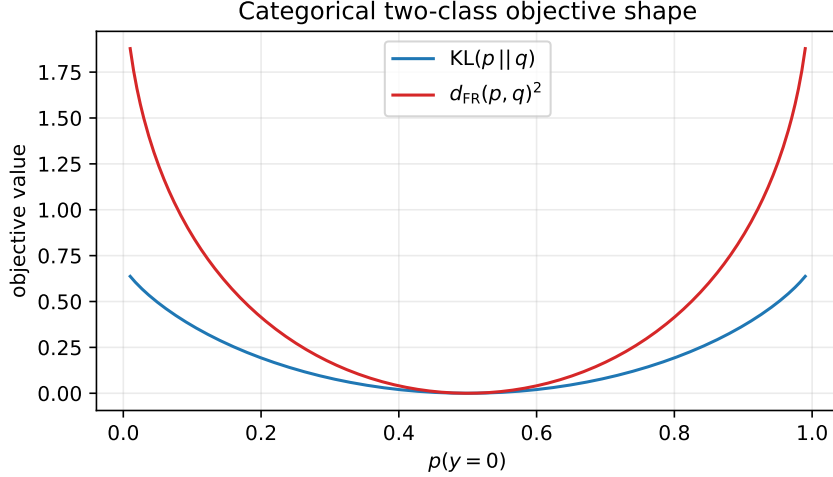


Figure 2: Two-class categorical objectives compared against a uniform target. The two curves agree to leading order near $p = q$ but differ globally: KL grows without bound while d_{FR}^2 saturates near π^2 .

3.3 Robustness corruption

We test robustness by injecting isotropic Gaussian noise into the high-dimensional features before constructing P . For a corruption level $\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$ measured in units of the post-standardization feature standard deviation,

$$\tilde{X}_\alpha = X + \alpha \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (8)$$

The embedding is optimized using $\tilde{X}_\alpha$ but evaluated against the clean labels and (for trustworthiness and neighborhood recall) against the clean X . This isolates the effect of the corruption on the affinity-building stage.

3.4 Evaluation metrics

The four embedding-quality metrics are objective-independent and standard in the neighborhood-embedding literature [9]:

- **Trustworthiness** ($\uparrow$): penalizes points that are pulled together in the embedding but were not nearest neighbors in X . Computed with $k = 10$.
- **Neighborhood recall** ($\uparrow$): fraction of X -space k -nearest neighbors recovered as Y -space k -nearest neighbors, $k = 10$.
- **Silhouette score** ($\uparrow$): mean silhouette of cluster labels in the embedding, measuring cluster separation in the embedding space.
- **kNN accuracy** ($\uparrow$): test accuracy of a 5-NN classifier on a 70/30 stratified split of the embedding, measured against the clean labels.

Table 1: Paired Wilcoxon signed-rank test of (Fisher–Rao – KL) across 10 seeds. Each cell shows the mean paired difference $\bar{\Delta}$ and Cliff’s delta δ (in brackets). * indicates $p < 0.05$. Negative $\bar{\Delta}$ means KL is better.

Dataset	Noise	Trust. ($\bar{\Delta}$ [δ])	Recall ($\bar{\Delta}$ [δ])	Silh. ($\bar{\Delta}$ [δ])	kNN ($\bar{\Delta}$ [δ])
Blobs	0.00	−0.002 [−0.06]	−0.002 [−0.54]	−0.017 [−0.22]	0.000 [0.00]
	0.25	−0.005 [−0.70]	−0.005 [−0.26]	−0.033 [−0.54]*	−0.005 [−0.21]
	0.50	−0.001 [0.10]	0.001 [0.07]	−0.003 [0.02]	−0.000 [0.01]
	0.75	−0.005 [−0.38]*	−0.004 [−0.15]	−0.033 [−0.54]*	−0.007 [−0.23]
	1.00	−0.014 [−0.58]*	−0.005 [−0.32]	−0.056 [−0.72]*	−0.042 [−0.31]*
Digits	0.00	−0.003 [−0.26]	−0.003 [−0.37]	−0.009 [−0.40]*	−0.015 [−0.28]
	0.25	0.002 [0.18]	0.001 [−0.10]	−0.007 [−0.40]*	0.020 [0.19]
	0.50	0.001 [0.00]	0.001 [−0.06]	−0.010 [−0.36]*	−0.017 [−0.06]
	0.75	−0.002 [−0.06]	0.005 [0.12]	−0.007 [−0.26]	0.043 [0.33]
	1.00	−0.002 [−0.08]	−0.002 [−0.10]	−0.014 [−0.42]*	0.000 [−0.01]
MNIST	0.00	0.002 [0.20]	−0.001 [−0.10]	−0.006 [−0.24]	−0.013 [−0.30]
	0.25	−0.000 [−0.14]	−0.002 [−0.17]	−0.006 [−0.30]	0.008 [0.06]
	0.50	0.002 [0.24]	−0.000 [−0.09]	−0.008 [−0.36]	−0.013 [−0.12]
	0.75	0.005 [0.36]	0.004 [0.13]	−0.003 [−0.24]	0.010 [0.08]
	1.00	−0.002 [−0.02]	−0.004 [−0.24]	−0.008 [−0.40]*	−0.007 [−0.05]

3.5 Statistical analysis

For each (dataset, noise) cell, the same 10 seeds are used to train both objectives. Each seed yields one paired observation $(x_i^{\text{KL}}, x_i^{\text{FR}})$ per metric. We report

- $x^{\text{KL}} \pm \text{std}$ and $x^{\text{FR}} \pm \text{std}$ across seeds,
- the paired difference $\Delta_i = x_i^{\text{FR}} - x_i^{\text{KL}}$, and
- the two-sided Wilcoxon signed-rank test [10] on Δ , which does not assume normality and handles small n .
- Cliff’s delta [2] as a non-parametric effect size, $\delta \in [-1, 1]$, with positive values meaning Fisher–Rao tends to exceed KL.

Following standard practice we mark $p < 0.05$ as significant. We do not apply Bonferroni across all 60 cells, but we report all p -values, so the reader can apply any preferred correction.

4 Results

Figure 3 shows mean ± 1 standard deviation across 10 seeds for each metric, dataset, and noise level. Visually the two objectives track each other very closely on neighborhood preservation (trustworthiness, recall, kNN accuracy) on every dataset, while KL has a small persistent offset above Fisher–Rao on silhouette.

4.1 Statistical significance

Table 1 reports the paired Wilcoxon signed-rank test and Cliff’s delta for each (dataset, noise) cell and each metric. Cells where the two objectives differ at $p < 0.05$ are marked with *.

The pattern is clean. Out of the 60 cells in Table 1,

- 13 have $p < 0.05$, and *all 13 favor KL*;

Final t-SNE embedding quality vs feature noise (mean $\pm$ std over seeds)

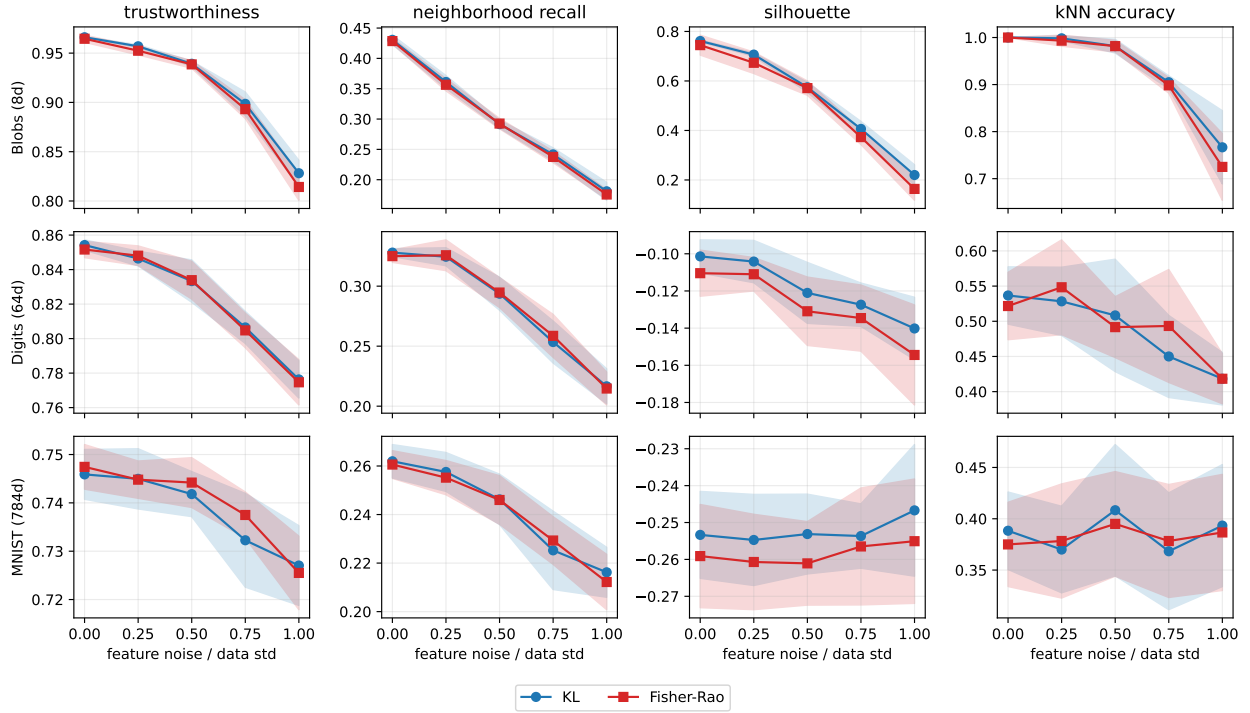


Figure 3: Final t-SNE embedding quality versus feature noise across three datasets, four evaluation metrics, two objectives, and ten seeds. Lines are seed means and shaded bands are ± 1 standard deviation. KL and Fisher-Rao track each other closely on neighborhood preservation but KL has a small consistent advantage on silhouette.

- 44 are statistically indistinguishable;
- the 13 KL-significant cells are concentrated on (i) silhouette, where KL wins on every dataset at multiple noise levels, and (ii) the simple Gaussian-blobs setting under strong corruption ($\alpha \in \{0.75, 1.0\}$);
- no cell shows a statistically significant Fisher-Rao advantage at the 0.05 level.

4.2 Per-metric average paired improvement

A complementary view averages the paired difference over noise levels for each (dataset, metric) pair (Figure 4). The picture is consistent with Table 1: silhouette is always negative, neighborhood preservation is near-zero with small dataset-dependent fluctuations, and the most favorable Fisher-Rao deviations are on digits and MNIST kNN accuracy at intermediate noise levels but do not survive the per-cell significance test.

4.3 Aggregated mean $\pm$ standard deviation

Table 2 reports per-cell mean and standard deviation values used in Figure 3, for the four metrics aggregated to a single number per metric per cell. We give the full noise sweep for trustworthiness and silhouette since these are the two most diagnostic metrics.

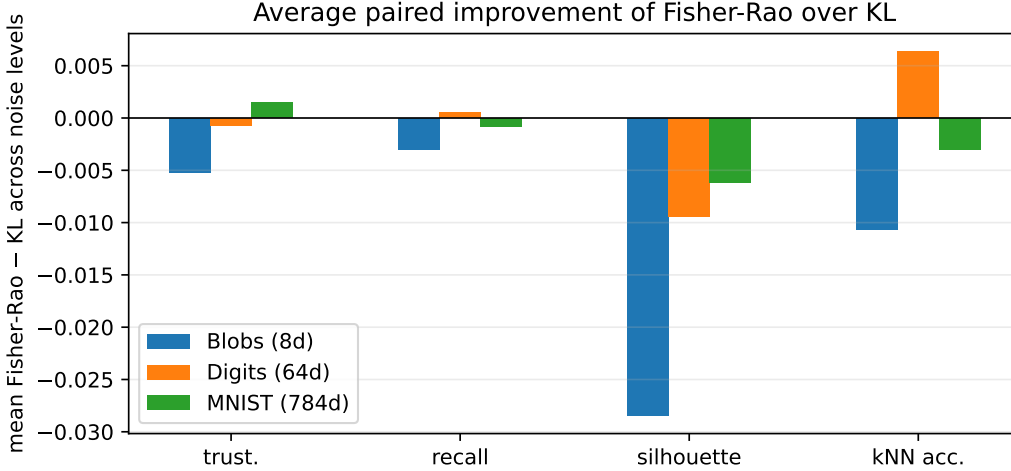


Figure 4: Average paired improvement of Fisher–Rao over KL for each metric, averaged across noise levels, broken down by dataset. Bars below zero mean KL is better on average.

4.4 Qualitative embeddings

Figure 5 shows the actual KL and Fisher–Rao embeddings on the **digits** dataset at three noise levels, with shared optimization seed and shared initialization. Cluster geometry is visually similar between objectives at every noise level; the silhouette gap captured in Tables 1–2 is small enough that it is not always obvious from the scatter plots.

4.5 Optimization dynamics

Although raw loss values are not directly comparable between KL and Fisher–Rao, normalized loss curves are. Figure 6 shows the normalized objective curves on **digits** at the moderate noise level, averaged across seeds. Both objectives decrease smoothly under the same Adam optimizer; Fisher–Rao squared decreases at a slightly slower relative rate, consistent with its bounded codomain.

5 Discussion

Why does KL win on cluster separation? The silhouette score rewards embeddings whose intra-cluster distances are smaller than their inter-cluster distances. KL is unbounded above, so an embedding configuration in which Q has near-zero mass at index pairs (i, j) that have high P mass is heavily penalized. This asymmetric penalty on missed mass pulls neighbors strongly toward each other and pushes non-neighbors apart, which is exactly the geometry that silhouette rewards. Fisher–Rao squared, in contrast, is capped at π^2 , so the penalty for misallocating mass cannot exceed this bound. The geometric advantage of bounded codomain becomes a practical disadvantage when the goal is aggressive cluster separation, because the objective stops responding once the geodesic angle reaches π .

Why are neighborhood metrics indistinguishable? KL and Fisher–Rao squared agree to leading order at $p = q$; they only differ globally. During optimization, especially in the late phase, Q approaches P for the high-probability pair indices, and the two objectives produce similar gradients

Table 2: Mean and standard deviation across 10 seeds for trustworthiness and silhouette, by dataset, noise level, and objective.

Dataset	Noise	Trustworthiness $\uparrow$		Silhouette $\uparrow$	
		KL	Fisher–Rao	KL	Fisher–Rao
Blobs	0.00	0.966 $\pm$ 0.000	0.965 $\pm$ 0.004	0.761 $\pm$ 0.003	0.745 $\pm$ 0.040
	0.25	0.957 $\pm$ 0.002	0.952 $\pm$ 0.004	0.706 $\pm$ 0.007	0.673 $\pm$ 0.043
	0.50	0.939 $\pm$ 0.003	0.939 $\pm$ 0.004	0.574 $\pm$ 0.020	0.571 $\pm$ 0.029
	0.75	0.899 $\pm$ 0.012	0.893 $\pm$ 0.009	0.406 $\pm$ 0.030	0.373 $\pm$ 0.030
	1.00	0.828 $\pm$ 0.013	0.814 $\pm$ 0.013	0.220 $\pm$ 0.041	0.164 $\pm$ 0.046
Digits	0.00	0.854 $\pm$ 0.003	0.852 $\pm$ 0.005	−0.101 $\pm$ 0.009	−0.110 $\pm$ 0.012
	0.25	0.846 $\pm$ 0.004	0.848 $\pm$ 0.006	−0.104 $\pm$ 0.011	−0.111 $\pm$ 0.009
	0.50	0.833 $\pm$ 0.012	0.834 $\pm$ 0.011	−0.121 $\pm$ 0.016	−0.131 $\pm$ 0.018
	0.75	0.806 $\pm$ 0.009	0.805 $\pm$ 0.010	−0.127 $\pm$ 0.012	−0.135 $\pm$ 0.018
	1.00	0.776 $\pm$ 0.011	0.775 $\pm$ 0.013	−0.140 $\pm$ 0.017	−0.154 $\pm$ 0.027
MNIST	0.00	0.746 $\pm$ 0.005	0.747 $\pm$ 0.005	−0.253 $\pm$ 0.012	−0.259 $\pm$ 0.014
	0.25	0.745 $\pm$ 0.006	0.745 $\pm$ 0.004	−0.255 $\pm$ 0.012	−0.261 $\pm$ 0.013
	0.50	0.742 $\pm$ 0.005	0.744 $\pm$ 0.005	−0.253 $\pm$ 0.011	−0.261 $\pm$ 0.011
	0.75	0.732 $\pm$ 0.010	0.737 $\pm$ 0.005	−0.254 $\pm$ 0.009	−0.256 $\pm$ 0.016
	1.00	0.727 $\pm$ 0.008	0.726 $\pm$ 0.008	−0.247 $\pm$ 0.018	−0.255 $\pm$ 0.017

there. The neighborhood metrics (trustworthiness, recall, kNN accuracy) measure agreement on these high-probability pairs, so they are close to insensitive to the global tail behavior that distinguishes the two objectives.

Why does KL especially dominate on Gaussian blobs at high noise? On well-separated isotropic clusters with strong noise, the clean affinity matrix P has a sparse block-diagonal structure: high mass within clusters, near-zero mass between clusters. Recovering this structure from a noisy $\hat{X}$ requires the objective to push between-cluster Q_{ij} aggressively toward zero. KL’s unbounded penalty on missed-mass deviations does this naturally; Fisher–Rao’s bounded penalty does not push as hard. In high-dimensional, less cleanly clustered settings (digits, MNIST), there is less sharp block structure to recover, so this advantage attenuates.

Where might Fisher–Rao still be the right tool? The bounded codomain that hurts t-SNE in our setting can help in other settings where KL’s unboundedness is precisely the problem. Two natural candidates are:

1. **Variational autoencoders with rare-event posteriors:** KL can dominate the objective when $q_\phi(z|x)$ assigns near-zero mass under the prior at some latent coordinates, leading to vanishing posterior collapse pressure or, at the other extreme, blow-ups during training. Fisher–Rao caps this contribution.
2. **Distribution matching with mass-mismatch:** when the two distributions being matched are known to have non-overlapping support (e.g., generative-model objectives comparing samples with different supports), KL is undefined or infinite while Fisher–Rao is finite.

We provide a small VAE preliminary in Appendix A that supports the first direction qualitatively, although a full VAE study is beyond the scope of this paper.

Methodological takeaway. The wider methodological point is that information-geometric replacements for KL should be evaluated with paired multi-seed protocols and explicit significance

t-SNE embeddings of digits dataset under increasing feature noise (single seed)

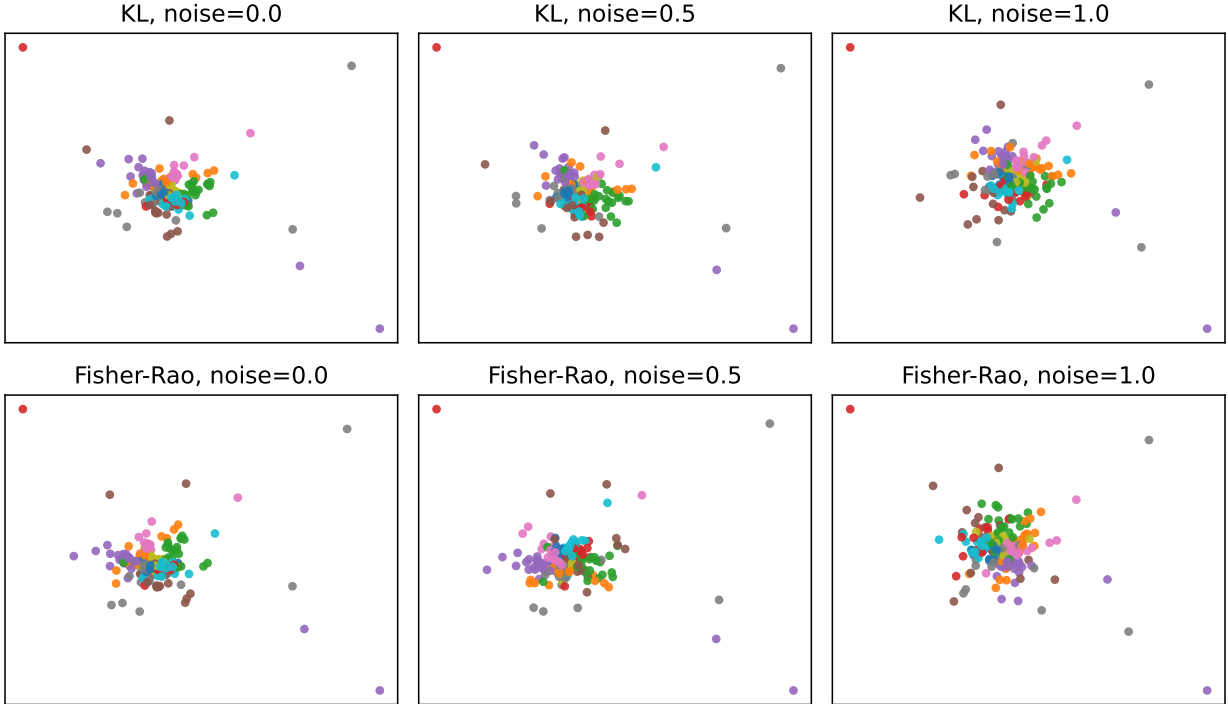


Figure 5: Qualitative t-SNE embeddings of `digits` (single seed) under three feature noise levels for the two objectives. The same initialization and optimizer are used; the two objectives produce visually similar embeddings.

tests rather than single runs. The infrastructure for doing so is small: our entire benchmark, including dataset loading, loss implementation, and statistical analysis, is < 500 lines of Python.

6 Limitations

The compact 200-point t-SNE size means the affinity matrix is 200×200 , well below practical t-SNE deployments at 10^4 – 10^5 points. Although nothing in the gradient code prevents scaling, the empirical patterns may shift for larger neighborhoods. We use a fixed median-distance bandwidth instead of perplexity-tuned bandwidths; this matches the simple synthetic baseline but is not the canonical t-SNE preprocessing. The robustness corruption is isotropic Gaussian noise; structured corruption (occlusions, label noise, manifold warping) might give a different ordering. Finally, we do not study early-exaggeration or annealing schedules, both of which interact non-trivially with objective scale. Each of these limitations defines a clean follow-up experiment in the same multi-seed framework.

7 Conclusion

The Fisher–Rao geodesic distance is a symmetric, triangle-inequality-respecting, parameterization-invariant, bounded-codomain alternative to KL divergence for comparing probability distributions.

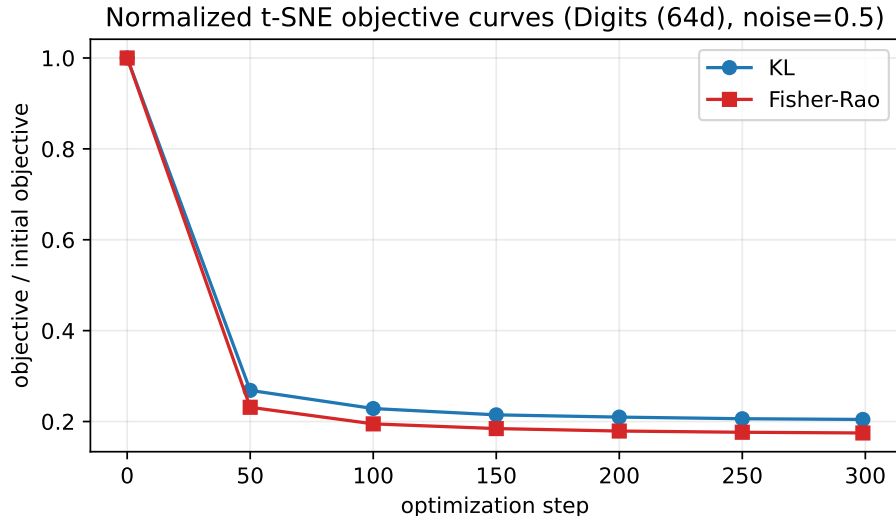


Figure 6: Normalized t-SNE objective curves on `digits` at noise $\alpha = 0.5$. Both objectives are well-behaved under Adam; the relative decrease rates differ because the underlying loss landscapes have different scales.

Despite this geometric appeal, when used as a t-SNE objective and evaluated with the same paired-comparison protocol that is now standard in empirical ML, it does not outperform KL: across 3 datasets, 5 corruption levels, and 10 seeds, KL significantly wins 13 out of 60 metric-cells (mostly silhouette and the strong-noise regime on simple Gaussian clusters), Fisher-Rao significantly wins zero, and the remaining 44 are statistical ties. Practitioners should continue to use KL for t-SNE-style affinity matching. The bounded-codomain property of Fisher-Rao remains attractive in settings where KL’s unboundedness is itself the problem, especially in VAE-like objectives where rare-event posteriors can destabilize the training signal; we leave a rigorous evaluation of those settings to follow-up work in the same framework.

A VAE Preliminary

We trained a small fully connected MNIST VAE with latent dimension 8, batch size 128, 2048 training points, 512 held-out points, Adam at learning rate 10^{-3} , gradient clipping at norm 10, for one epoch, with three regularizers: KL at $\beta = 1$, Fisher-Rao squared at $\beta = 1$, and Fisher-Rao squared at $\beta = 0.3$. This is a single-seed preliminary, not a paired benchmark, and is included here only to give a directional sense of the trade-offs.

Table 3: VAE preliminary on MNIST, single seed.

Reg.	β	BCE/pix $\downarrow$	Noisy BCE $\downarrow$	Mean norm	Var. mean	Active	Latent kNN $\uparrow$
KL	1.0	0.290	0.286	5.59	1.75	7	0.439
Fisher-Rao	1.0	0.293	0.291	5.59	2.66	8	0.455
Fisher-Rao	0.3	0.292	0.294	7.12	2.45	8	0.479

The directional pattern is that Fisher-Rao activates more latent units (8 vs 7) and yields slightly better latent-mean kNN accuracy, at the cost of marginally higher reconstruction error. Lowering β for Fisher-Rao further improves latent kNN accuracy but slightly degrades noisy-input

reconstruction. A proper multi-seed VAE benchmark would be needed to verify whether this is a real effect.

B Dataset and Corruption Details

B.1 Blobs

Generated with `sklearn.datasets.make_blobs(n_samples=200, n_features=8, centers=5, cluster_std=1.6, random_state=0)`, then z-scored. Five Gaussian centers in $\mathbb{R}^8$.

B.2 Digits

A 200-point random subsample (numpy default RNG seeded 0) from sklearn’s `load_digits` bundle, which contains 1797 images of 8×8 pixel handwritten digits, 10 classes, and 64 raw features. Z-scored to zero mean and unit variance.

B.3 MNIST

A 200-point random subsample (numpy default RNG seeded 0) from the MNIST training split, each image flattened to 784 raw pixel features and z-scored.

B.4 Robustness corruption

For each noise level $\alpha \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ and each seed $s \in \{101, 202, \dots, 1010\}$, we draw a fresh standard normal noise tensor of the same shape as the standardized feature matrix and form $\tilde{X} = X + \alpha \epsilon$. The seed for the noise tensor is $1000s + \lfloor 1000\alpha \rfloor$ to keep noise shocks reproducible and distinct across cells.

C Training Dynamics Details

All t-SNE benchmark runs use Adam at learning rate 0.05 for 300 steps, with gradient descent on the 200×2 embedding matrix. Optimizer state is reset for each (dataset, noise, seed, objective) run. The high-dimensional bandwidth is set per dataset to $\sigma = \text{median}_{i < j} \|x_i - x_j\| / \sqrt{2}$ on the clean standardized data, giving $\sigma \approx 2.81$ for blobs, $\sigma \approx 7.0$ for digits, and $\sigma \approx 21.8$ for MNIST.

The full benchmark, including aggregation and statistical analysis, runs in approximately two and a half minutes on Apple Silicon MPS using the script `experiments/paper_benchmark.py` followed by `experiments/aggregate_results.py`, with figures regenerated by `reports/generate_figures.py`. End-to-end the paper can be rebuilt with `make paper-all`.

References

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